

Chapter 1

INTRODUCTION TO ALGORITHMS AND FLOWCHARTS

Chapter 1

- **Algorithms**
- **Using Flowchart to describe Algorithm**
- **Algorithm in pseudo-code**
- **Branching in Algorithms**
- **Loop in Algorithms**

1. Algorithms

- An ***algorithm*** is defined as a step-by-step sequence of instructions that describes how the data are to be processed to produce the desired outputs.
- In essence, an algorithm answers the question: “What method will you use to solve the problem?”.
- You can describe an algorithm by using flowchart symbols. By that way, you obtain a flowchart.
- ***Flow chart*** is an outline of the basic structure or logic of the program.
- Another way to describe an algorithm is using ***pseudocode***.

2. Using Flowchart to describe algorithm

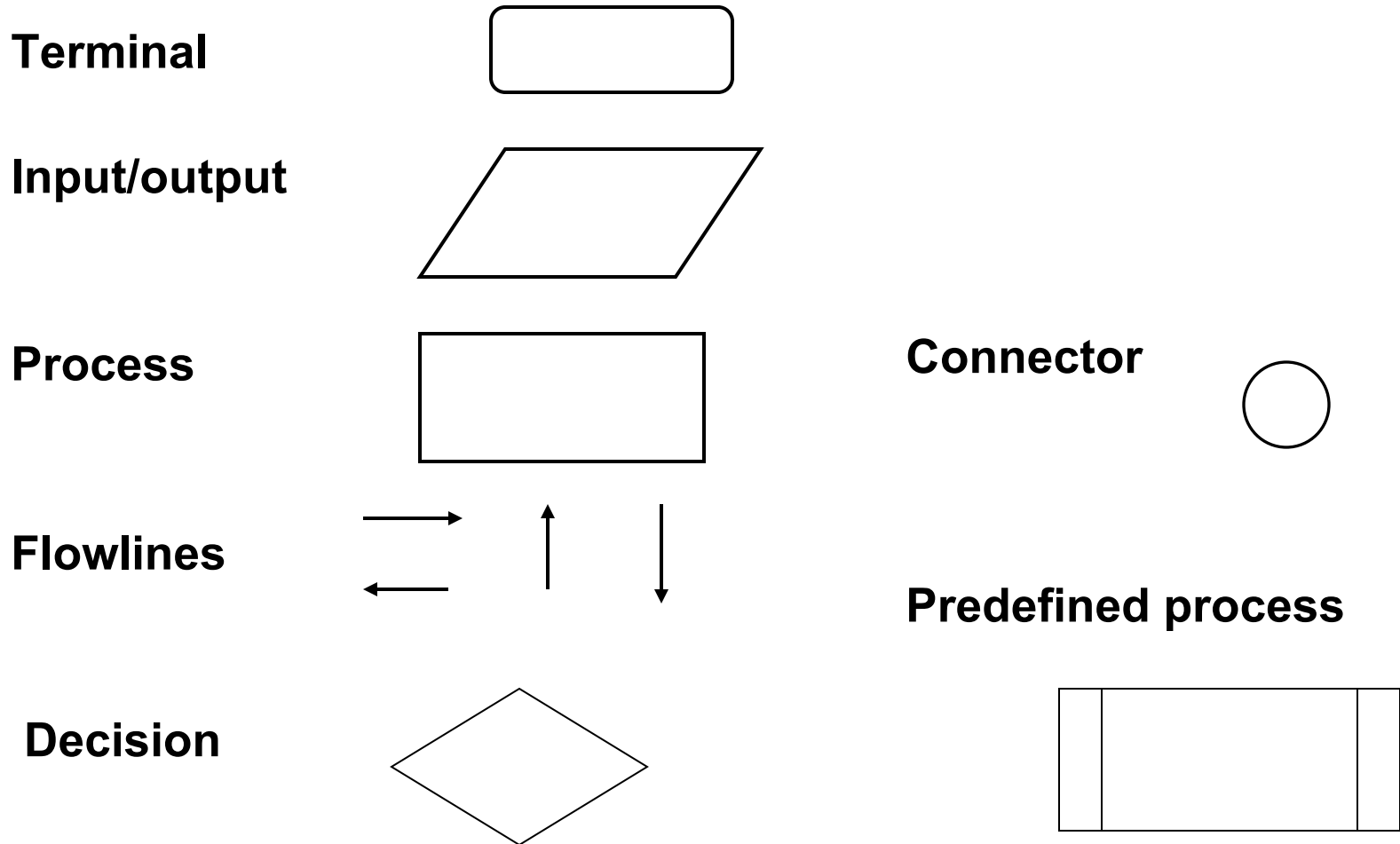
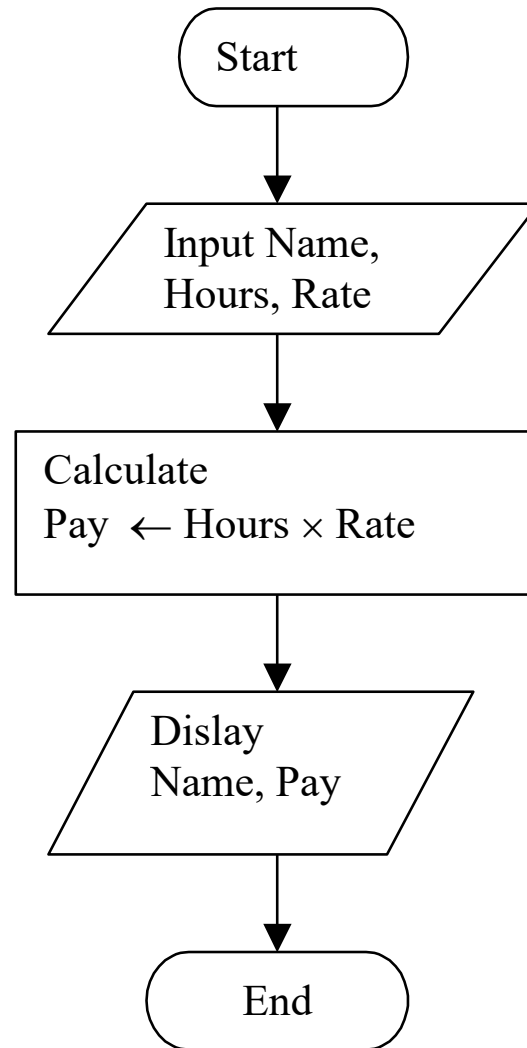


Fig 1. Flowchart symbols

Example 2.1. Calculate the payment of an employee

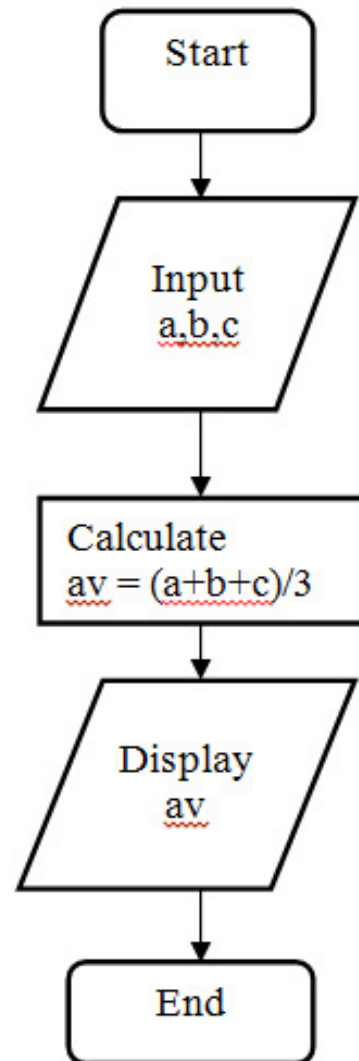
Fig. 2.



Note: Name, Hours and Pay are *variables* in the program.

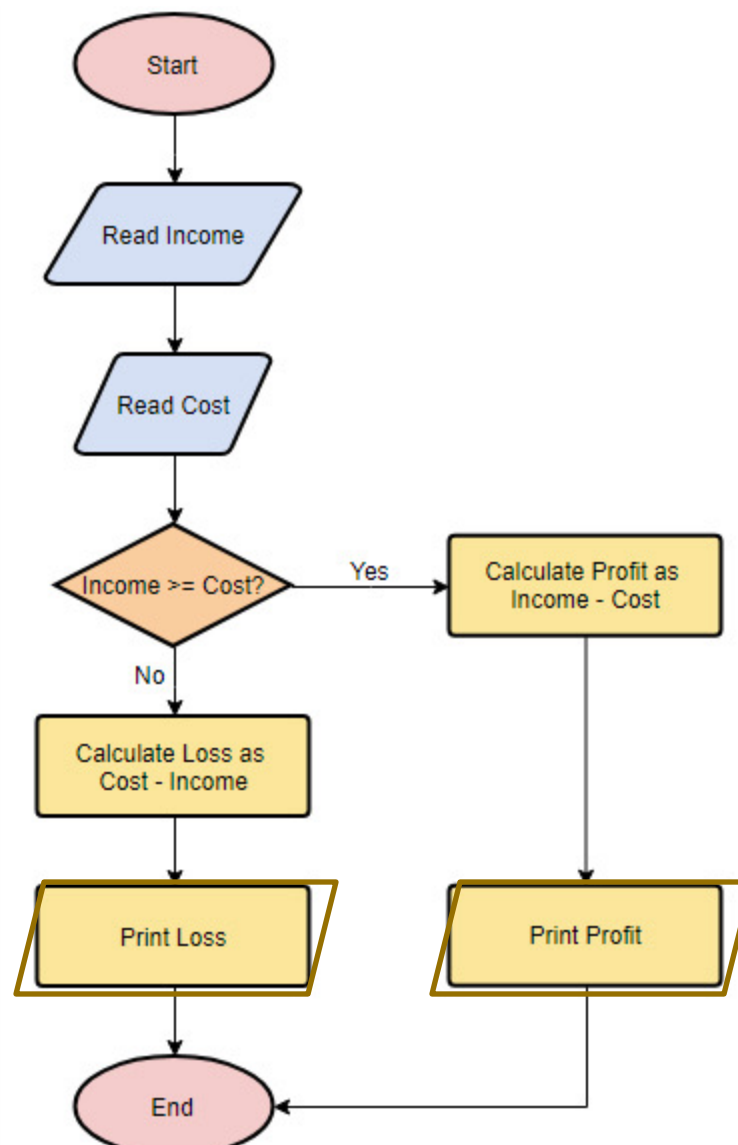
Example 2.2: Calculate the average of 3 numbers

Fig. 3.



Example 2.3: How to compute profit and loss.

Fig. 4.



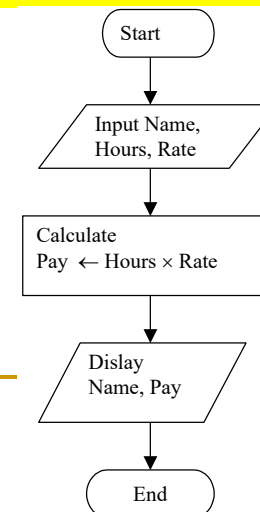
3. Algorithms in pseudo-code

- You also can use English-like phrases to describe an algorithm. In this case, the description is called *pseudocode*.
- Example: The pseudocode for the program with the flowchart in Fig. 2.

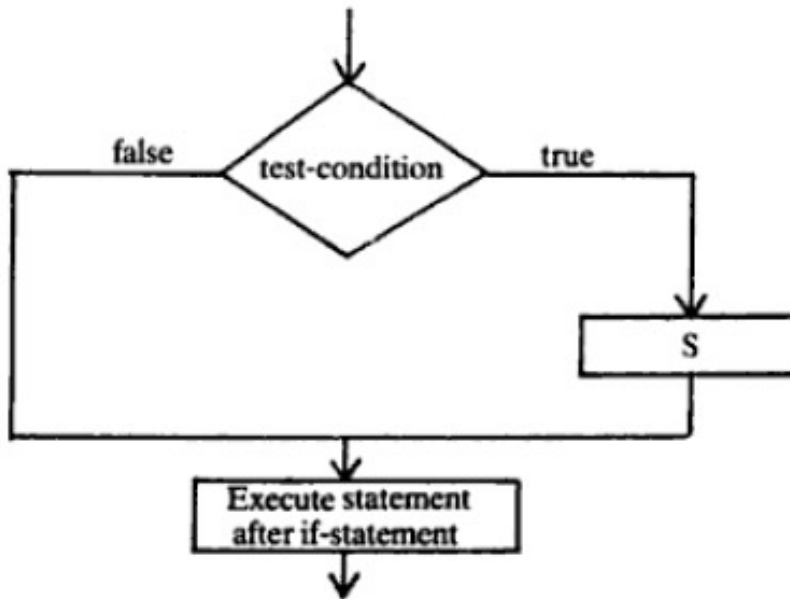
Input the three values into the variables Name, Hours, Rate.

Calculate $Pay = Hours \times Rate$.

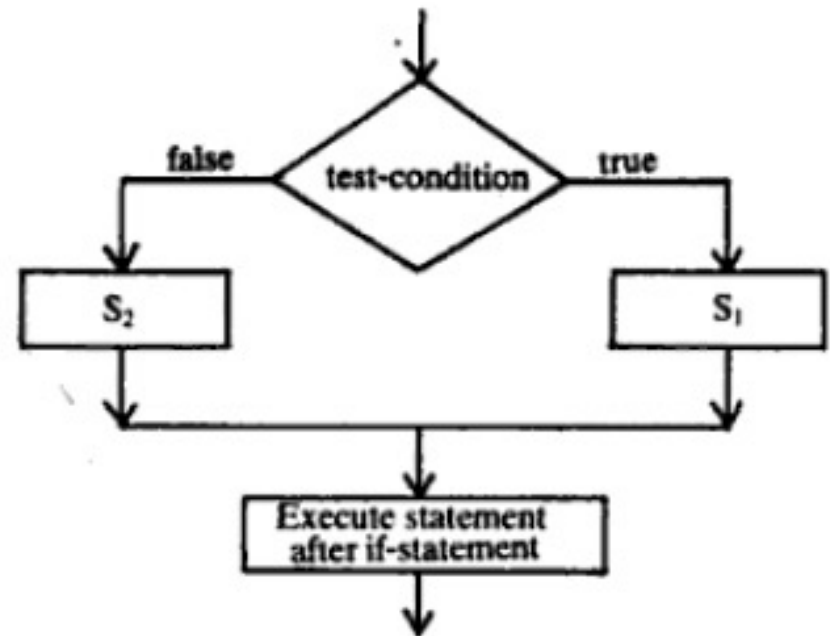
Display Name and Pay.



4. Branching in algorithms



**If test-condition then S
endif**



**If test-condition then S₁
else S₂
endif**

Example 2.1

Pseudo-code of the program:

Input A

if $A > 0$ **then**

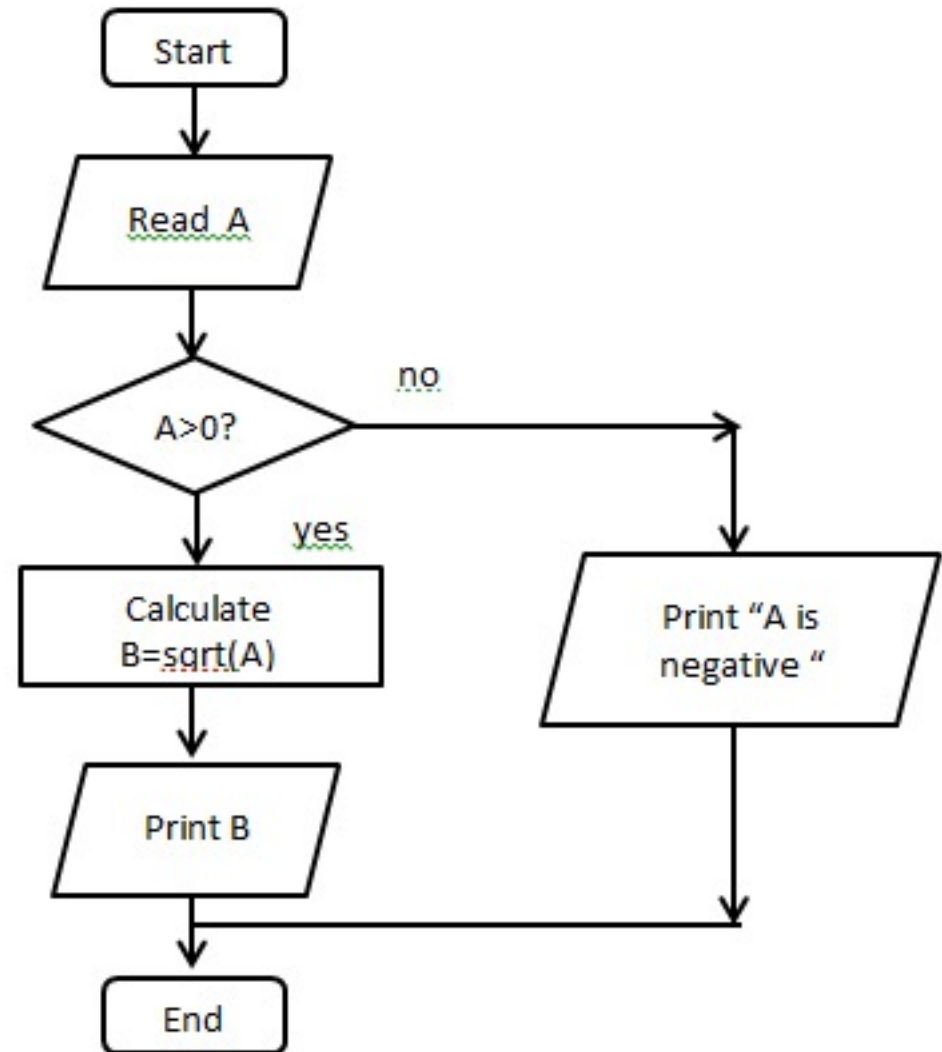
 calculate $B = \text{sqrt}(A)$

 print B

else

 Print "A is negative"

endif



Branching

Example 4.2: Solving the quadratic equation:
 $ax^2 + bx + c = 0$

Pseudo-code of the program:

Input a, b, c

Calculate $\text{del} = b^2 + 4.a.c$

if $\text{del} > 0$ **then**

$x1 = (-b + \text{sqrt}(\text{del})) / (2.a)$

$x2 = (-b - \text{sqrt}(\text{del})) / (2.a)$

 print x1, x2

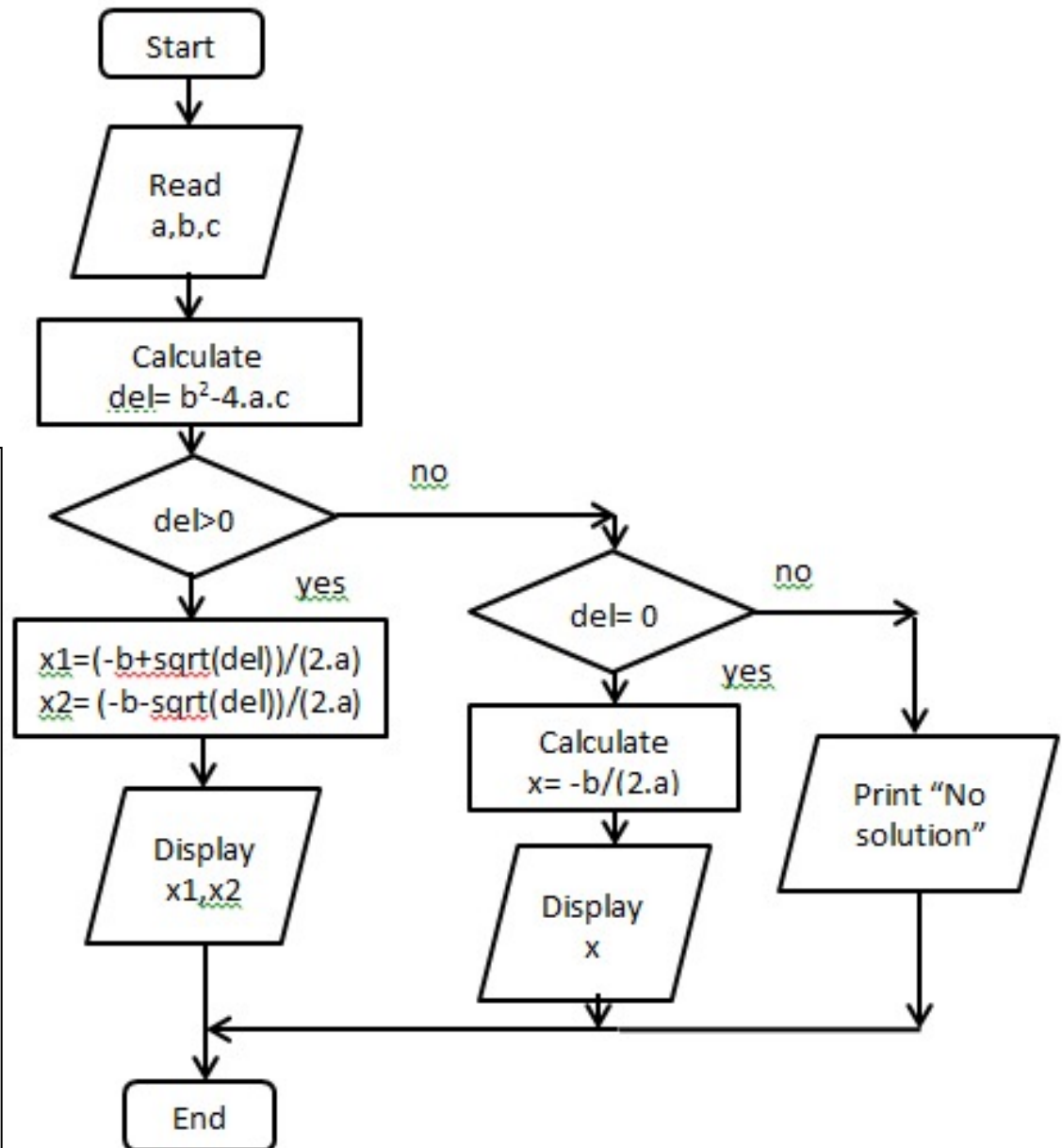
else if $\text{del} = 0$ **then**

$x = -b / (2.a)$

else

 print "No solution"

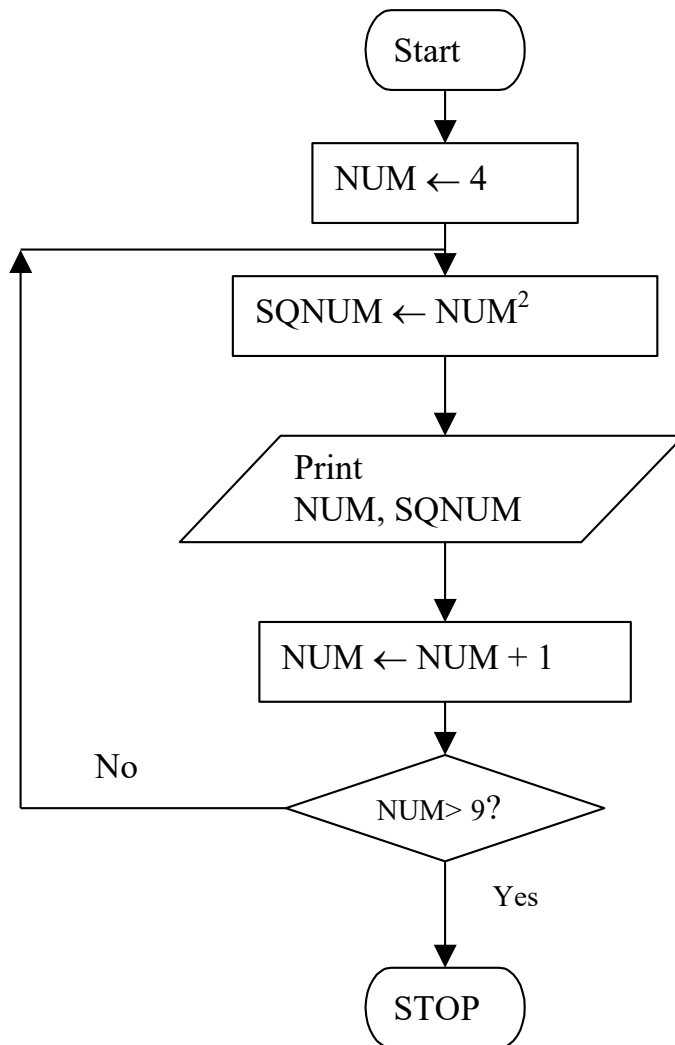
endif



5. Loops on Algorithms

- Many problems require repetition capability, in which the same calculation or sequence of instructions is *repeated*, over and over, using different sets of data.
- Loop is a very important concept in programming.
- Example 5.1. Write a program to do the task: Print a list of the numbers from 4 to 9, next to each number, print the square of the number.

Example 5.1



Note:

**$NUM \leftarrow NUM + 1$ means
old value of $NUM + 1$
becomes new value of NUM .**

**The algorithm can be described
in pseudocode as follows:**

$NUM \leftarrow 4$

do

$SQNUM \leftarrow NUM^2$

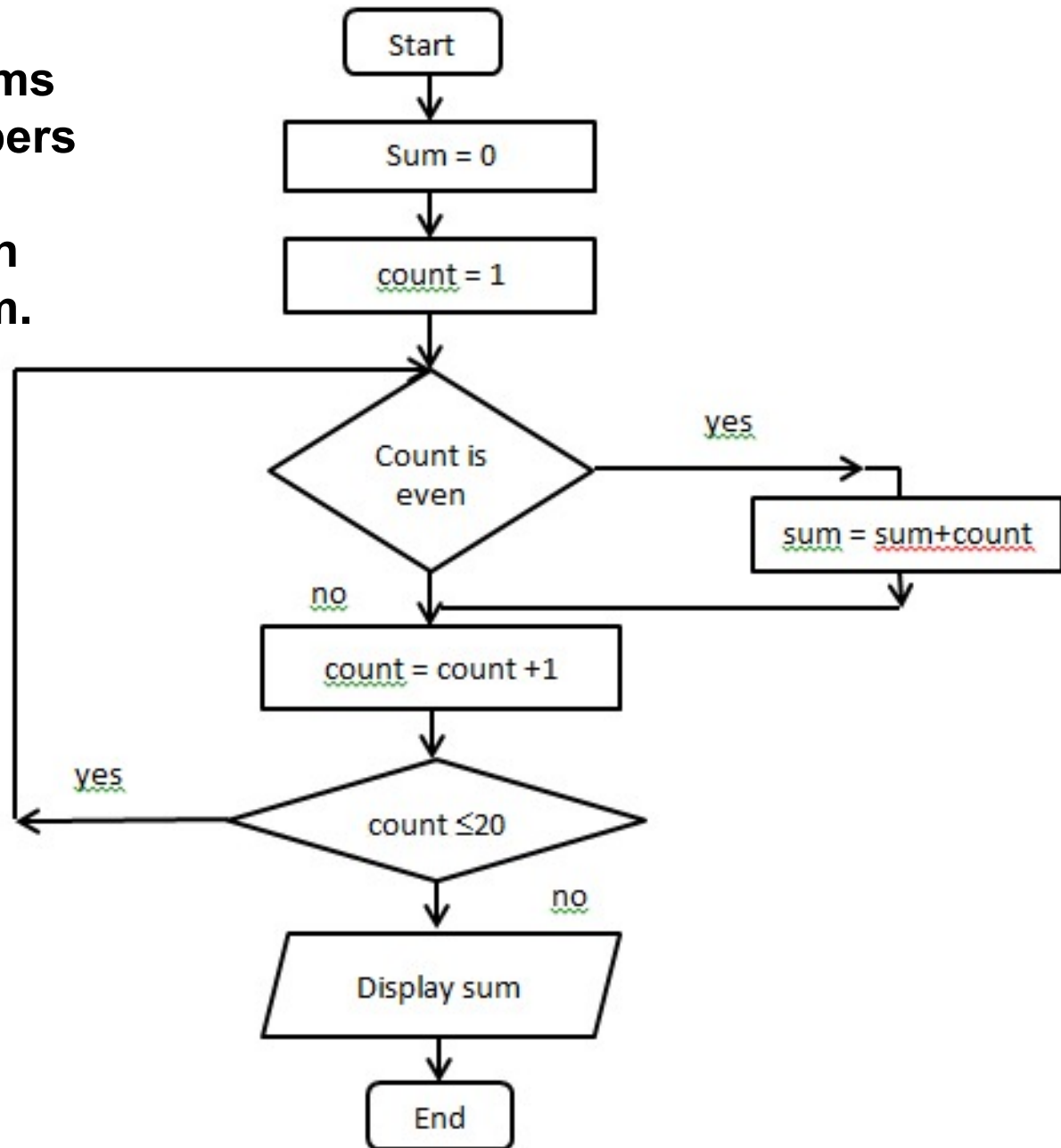
Print $NUM, SQNUM$

$NUM \leftarrow NUM + 1$

while ($NUM \leq 9$)

Example 5.2

The algorithm sums all the even numbers between 1 and 20 inclusive and then displays then sum.



Pseudo-code of Example 5.2

sum = 0

count = 1

do

if count is even then

 sum = sum + count

endif

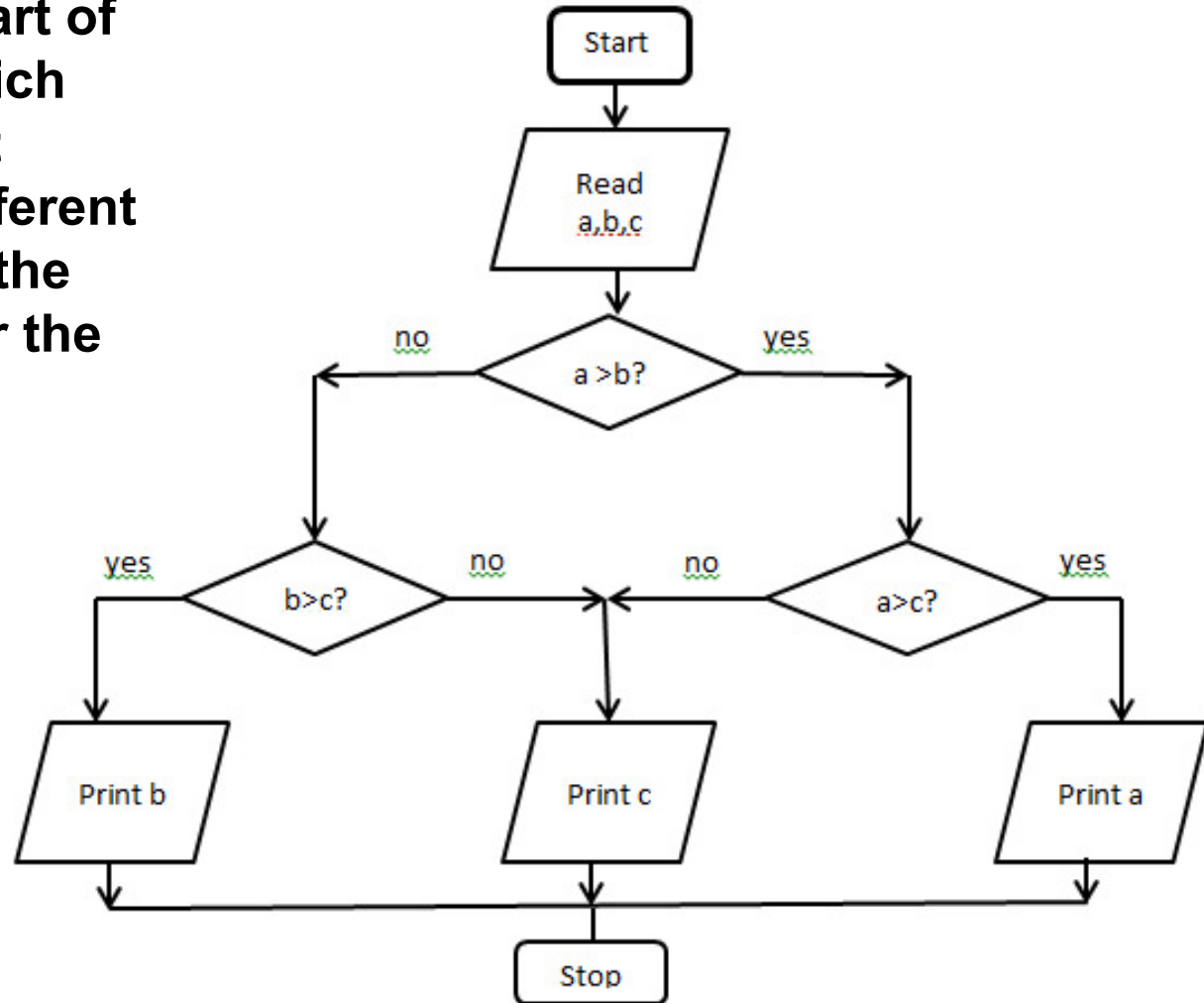
 count = count + 1

while count <= 20

Display sum

Exercise

Give the flowchart of the program which finds the largest among three different numbers. Write the pseudo-code for the flowchart.



Ans.

Input a,b,c

if a > b then

if a > c then

 print a

endif

else

if b > c then

 print b

else

 print c

endif

endif